MTH636 – Game Theory

Lecture 17: Subgames & Zermelo's Theorem

1 Strategy Profiles and Paths

In an extensive form game $\Gamma = \langle N, V, E, n_0, (V_i)_{i\in N}, (u_i)_{i\in N}\rangle$, a strategy for player i is a map $s_i : V_i \to \cup A(n)$.

A **strategy profile** $s = (s_1, \ldots, s_n)$ is a tuple containing a strategy for each player. Every strategy profile uniquely determines a specific path in the game tree, starting from the root n_0 and ending at a leaf node n_k:

$$(n_0, n_1, \ldots, n_k)$$

The utility of the strategy profile is simply defined as the utility at the leaf node reached:

$$u(s) := u(n_k)$$

2 Subgames

Definition: Subgame

Let Γ be an extensive form game. Consider a node $n \in V$. The **subgame starting at** n, denoted by $\Gamma(n)$, is an extensive form game itself:

$$\Gamma(n) = \langle N, V(n), E(n), n, (V_i(n))_{i\in N}, (u_i)_{i\in N}\rangle$$

where:

- N: The set of players remains the same.
- $V(n)$: The set of all nodes that can be reached from n (descendants of n, including n).
- $E(n)$: All the edges that connect nodes within $V(n)$.
- n: The new root of the subgame.
- $V_i(n) = V_i \cap V(n)$: The nodes controlled by player i within the subgame.
- u_i: The restriction of the utility function u_i to the leaves in the new game tree $(V(n), E(n), n)$.

Remarks:

- The original game is a subgame of itself: $\Gamma = \Gamma(n_0)$.
- Every leaf node trivially defines a subgame where no player makes a move, and payoffs are immediately realized.

3 Win/Loss/Draw Games & Zermelo's Theorem

Consider a particular type of game:

- Two players.
- Perfect information.
- Finite duration.
- Only three possible outcomes:
  1. Player 1 wins (payoffs $1, -1$)
  2. Player 2 wins (payoffs $-1, 1$)
  3. The game ends in a draw (payoffs $0, 0$)

Examples include Tic-Tac-Toe, Checkers, and Chess (ignoring infinite loops).

Definition: Winning and Drawing Strategies

- A **winning strategy** for Player 1 is a strategy s_1 such that $u(s_1, s_2) = (1, -1)$ for *all* possible opponent strategies $s_2 \in S_2$.
- A **strategy guaranteeing at least a draw** for Player 1 is a strategy s_1 such that $u(s_1, s_2) \in \{(1, -1), (0, 0)\}$ for all $s_2 \in S_2$.

(Analogous definitions apply for Player 2).

Theorem: Zermelo's Theorem (1913) / von Neumann (1928)

In every two-player, finite game of perfect information in which the set of utilities is $\{(1, -1), (-1, 1), (0, 0)\}$, **one and only one** of the following holds:

1. Player 1 has a winning strategy.
2. Player 2 has a winning strategy.
3. Both players have strategies guaranteeing at least a draw.

4 Introduction to Chomp (David Gale, 1974)

To illustrate these concepts, we introduce the game **Chomp**.

- Played on an $m \times n$ grid of squares (like a chocolate bar).
- The bottom-left square is "poisoned."
- Players take turns choosing a valid square (i, j).
- When a square is chosen, that square and all squares above and to the right of it are "eaten" and removed from the board.
- The player forced to eat the poisoned bottom-left square loses.

Because Chomp is finite, has perfect information, and has no draws, Zermelo's theorem guarantees that either Player 1 or Player 2 must have a winning strategy. We will prove who wins in the next lecture.